# Practice workbook · Pixels, resolution and colour depth

Concepts of bitmap representation: what a pixel is, measures of resolution, and why more colour depth makes a bigger file.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-10

# 1. Question 3 · part b.ii · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

**3** Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

**(2)**

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

**(1)**

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

**(2)**

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# 1. Question 3 · part b.ii · 2 marks

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Worked-example retry · Attempt part b.ii only.

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(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

**(2)**

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | W4 | Z1 | | | | W2 | | W3 | Z3 | | W1 | | Z2 |

Complete the image to show the state after running a defragmentation utility.

**(2)**

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | |

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P 7 2 9 3 7 A 0 9 2 4

## 1. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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## 2. Question 1 · part g.i · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part g.i only.

**1** Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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## 2. Question 1 · part g.i · 1 mark

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Further practice · Attempt part g.i only.

(d) Complete the table by adding these two 8-bit binary integers.

**(2)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| | | | | | | | |

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

**(1)**

.....

(f) Give the number of bits per character used by standard ASCII.

**(1)**

.....

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

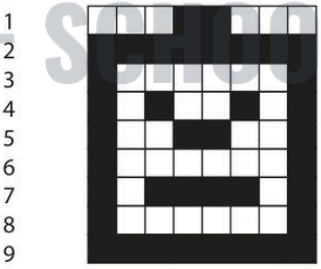



**Figure 1**

(i) State what is meant by the term **pixel**.

**(1)**

.....

(ii) Construct an expression to show the number of pixels in the image.

**(1)**

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## 2. Question 1 · part g.i · 1 mark · Working and answer

Working and answer

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# 3. Question 3 · part b.i · 1 mark

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Further practice · Attempt part b.i only.

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(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

**(2)**

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

**(1)**

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

**(2)**

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**8**

P 7 2 9 3 7 A 0 8 2 4

## 3. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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# 4. Question 3 · part b.i · 1 mark

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Further practice · Attempt part b.i only.

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**3** Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

**(2)**

(ii) Identify the statement that is true for hexadecimal.

**(1)**

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

**(2)**

# 4. Question 3 · part b.i · 1 mark

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Further practice · Attempt part b.i only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

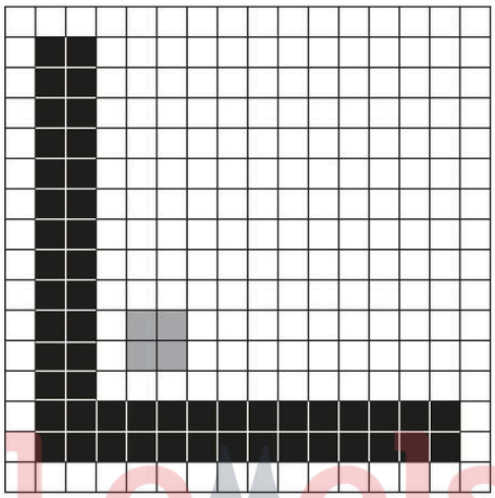


**Figure 1**

(i) State the smallest component of a bitmap image.

**(1)**

.....

.....

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

**(4)**

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P 7 8 6 3 7 A 0 8 2 0

## 4. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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